

DG Interns Hub – Cybersecurity Internship



Week 3 Assignment Report: Network Security

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INTRODUCTION

- This practical assignment focuses on performing an authorized cybersecurity assessment in a controlled virtual lab environment. The assessment uses Kali Linux and a Windows 7 test virtual machine to study network discovery, service enumeration, vulnerability assessment, and network traffic analysis.
- The objective of the final assessment was to evaluate the security posture of the authorized Windows 7 test VM after performing network discovery, Nmap scanning, Wireshark traffic analysis, vulnerability assessment, and security hardening.

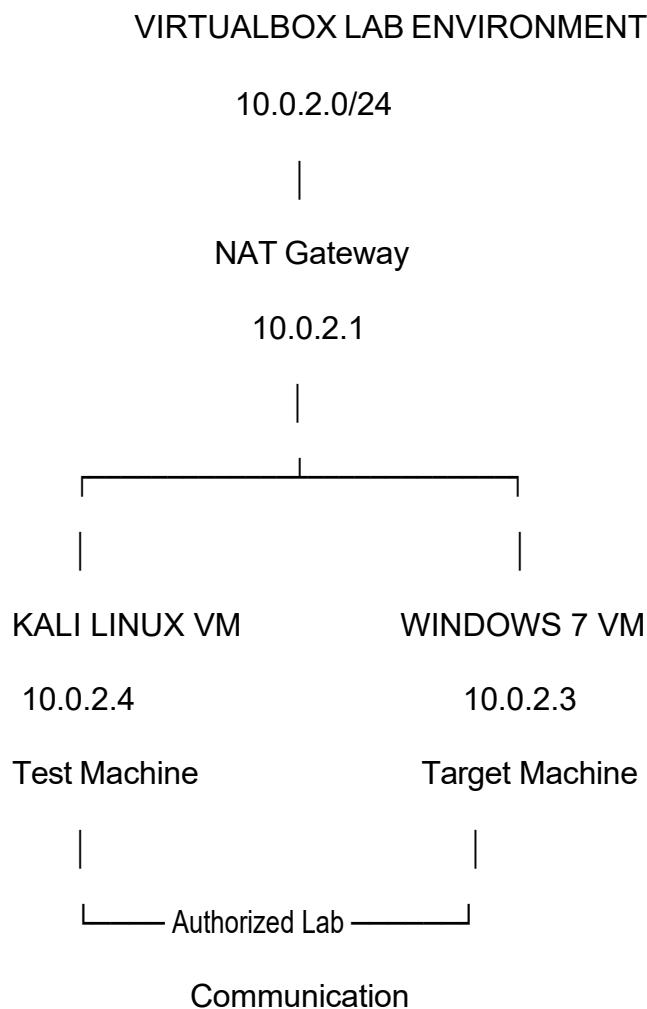
Objectives

- **To perform network discovery** using Nmap and identify active hosts within the lab network.
- **To identify open ports and services** running on the Windows 7 test VM using Nmap scanning techniques.
- **To capture and analyze network traffic** using Wireshark and study protocols such as DNS, TCP, UDP, ARP, and ICMP.
- **To correlate Nmap results with Wireshark traffic** and understand how network scanning generates TCP packets such as SYN, SYN/ACK, and RST.
- Apply at least three concrete hardening measures and verify the resulting reduction in attack surface.
- • Produce a professional, evidence-backed security assessment and a set of actionable recommendations.

Lab Environment

The practical cybersecurity assessment was conducted in a **controlled virtual laboratory environment** created using VirtualBox. The architecture consists of two virtual machines: a **Kali Linux VM** used for security testing and a **Windows 7 VM** used as the authorized target system.

Network Diagram



TOOLS USED

Tool	Purpose
Kali Linux	Security testing and assessment environment
Windows 7 VM	Authorized target/test system
VirtualBox	Created and managed the virtual lab environment
Nmap	Host discovery, port scanning, service/version detection, OS detection and vulnerability assessment
Wireshark	Network traffic capture and packet analysis
Windows Firewall	Applied security hardening and restricted inbound ports
diagrams.net (draw.io)	Created the network topology diagram



TASK - 7 Network Discovery & Basic Nmap

Network Discovery : Network discovery is the process of identifying active devices within a network. It helps a security professional understand which hosts are reachable and which systems may need further security assessment

- **Network:** 10.0.2.0/24
- **Kali Linux:** 10.0.2.4
- **Windows 7:** 10.0.2.3

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:34:fe:ef brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.4/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 303sec preferred_lft 303sec  
    inet6 fe80::541c:c6cd:25a1:a94a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)-[~]  
$
```

```
C:\Users\ vboxuser>ipconfig  
  
Windows IP Configuration  
  
Ethernet adapter Local Area Connection:  
  
    Connection-specific DNS Suffix  . : domain.name  
    Link-local IPv6 Address . . . . . : fe80::a9d6:d860:16ab:4a7a%11  
    IPv4 Address. . . . . : 10.0.2.3  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 10.0.2.1  
  
Tunnel adapter isatap.domain.name:  
  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . : domain.name  
  
C:\Users\ vboxuser>ping 10.0.2.4  
  
Pinging 10.0.2.4 with 32 bytes of data:  
Reply from 10.0.2.4: bytes=32 time=1ms TTL=64  
Reply from 10.0.2.4: bytes=32 time<1ms TTL=64  
Reply from 10.0.2.4: bytes=32 time<1ms TTL=64  
Reply from 10.0.2.4: bytes=32 time<1ms TTL=64  
  
Ping statistics for 10.0.2.4:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 0ms, Maximum = 1ms, Average = 0ms  
  
C:\Users\ vboxuser>
```

Nmap (Network Mapper) is a network security tool used to discover hosts and identify open ports and services running on authorized systems. It is commonly used during network security assessments and vulnerability investigations.

```
(kali@kali)~$ nmap -sn 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-19 13:38 -0400
Stats: 0/0/0 elapsed; 0 hosts completed (2 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 91.00% done; ETC: 13:41 (0:00:18 remaining)
Nmap scan report for 192.168.1.10
Host is up.
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 202.09 seconds
```

nmap -sn 10.0.2.0/24 this command performs host discovery across the specified network. It determines which hosts are responding without performing a normal port scan.

```
(kali@kali)~$ nmap -sn 10.0.2.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-19 15:39 -0400
Nmap scan report for 10.0.2.1
Host is up (0.00052s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.2
Host is up (0.00031s latency).
MAC Address: 08:00:27:6A:85:C7 (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.3
Host is up (0.00081s latency).
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.4
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.00 seconds

(kali@kali)~$
```

nmap 10.0.2.3 scans the authorized Windows 7 VM for commonly used TCP ports.

```
(kali@kali)~$ nmap 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-19 15:48 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00064s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
49157/tcp  open  unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds

(kali@kali)~$
```



Task 8 – Advanced Nmap Security Assessment

Service and Version Detection : nmap -sV 10.0.2.3 The `-sV` option attempts to determine the service and version associated with open ports.

```
Session Actions Edit View Help

(kali@kali)-[~]
$ nmap -sV 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-19 16:23 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0019s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
49157/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)
Service Info: Host: WINDOWS7; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 61.03 seconds
```

OPERATING SYSTEM DETECTION : sudo nmap -O 10.0.2.3 The `-O` option performs operating-system detection by analyzing characteristics of the target's network responses.

```
Session Actions Edit View Help

(kali@kali)-[~]
$ sudo nmap -O 10.0.2.3
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-19 16:26 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0019s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.84 seconds
```

TCP SYN SCAN : `sudo nmap -sS 10.0.2.3` a SYN scan examines TCP ports by sending SYN packets and analyzing the responses. It is commonly used to identify accessible TCP services during a security assessment.

```
(kali@kali)-[~]
└─$ sudo nmap -sS 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-19 16:27 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0026s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
49157/tcp  open  unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.83 seconds
```

UDP SCAN : `sudo nmap -sU --top-ports 20 10.0.2.3` The `-sU` option scans for UDP services, while `--top-ports 20` limits the scan to the 20 most commonly used UDP ports.

```
(kali@kali)-[~]
└─$ sudo nmap -sU --top-ports 20 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-19 16:28 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0012s latency).

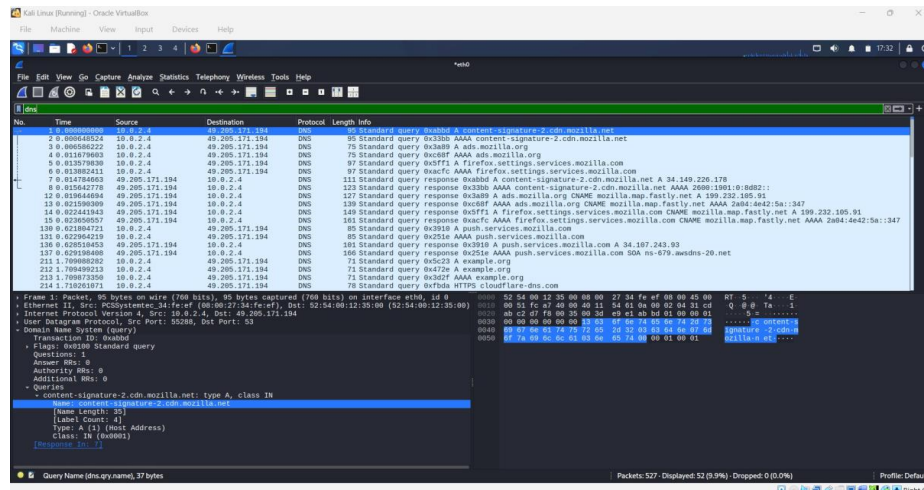
PORT      STATE SERVICE
53/udp    open|filtered domain
67/udp    open|filtered dhcps
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
123/udp   open|filtered ntp
135/udp   open|filtered msrpc
137/udp   open      netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   closed    netbios-ssn
161/udp   open|filtered snmp
162/udp   open|filtered snmptrap
445/udp   open|filtered microsoft-ds
500/udp   open|filtered isakmp
514/udp   open|filtered syslog
520/udp   open|filtered route
631/udp   open|filtered ipp
1434/udp  open|filtered ms-sql-m
1900/udp  open|filtered upnp
4500/udp  open|filtered nat-t-ike
49152/udp open|filtered unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.04 seconds
```

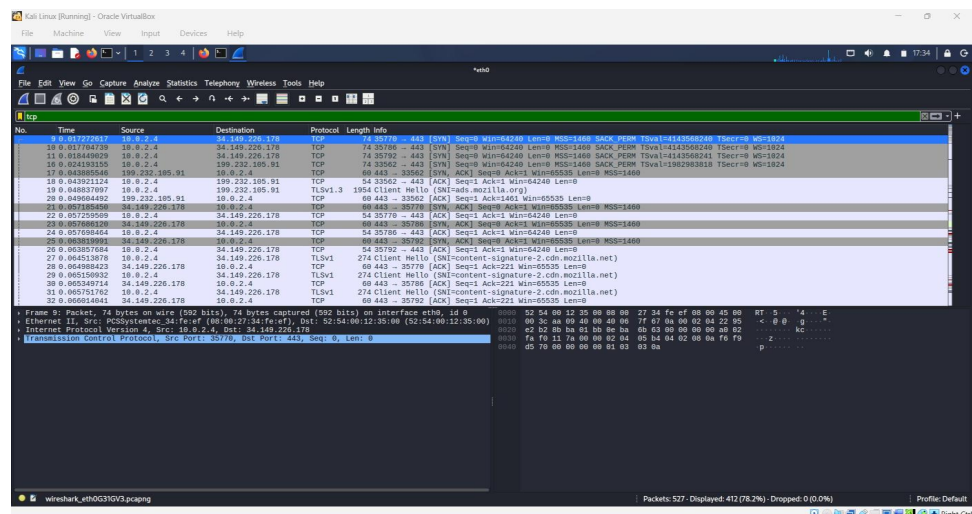


Task 9 – Wireshark Network Traffic Capture

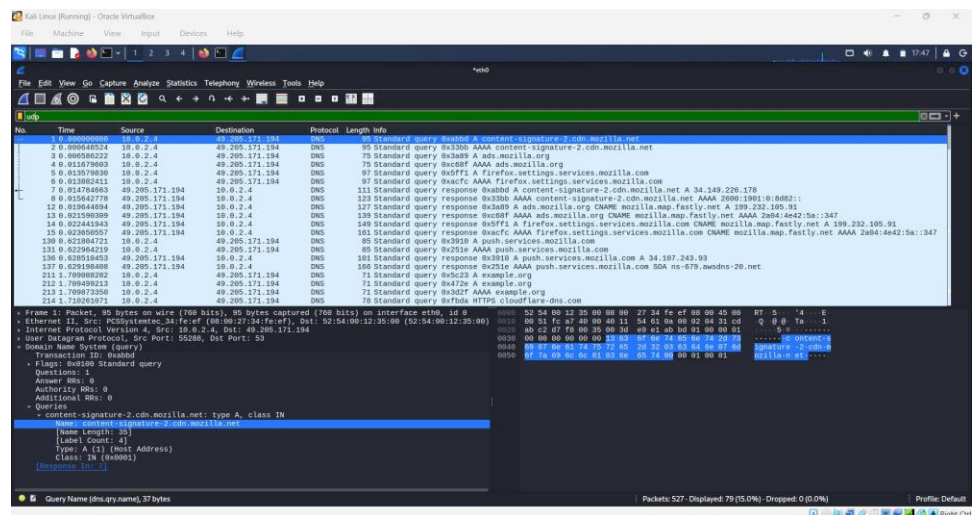
DNS : Capturing Wireshark running on a Kali Linux virtual machine with the display filter set to `dns`, effectively isolating Domain Name System query and response traffic over the `eth0` network interface. The traffic log clearly displays interaction between the client workstation at IP `10.0.2.4` and the upstream DNS resolver at `49.205.171.194`.



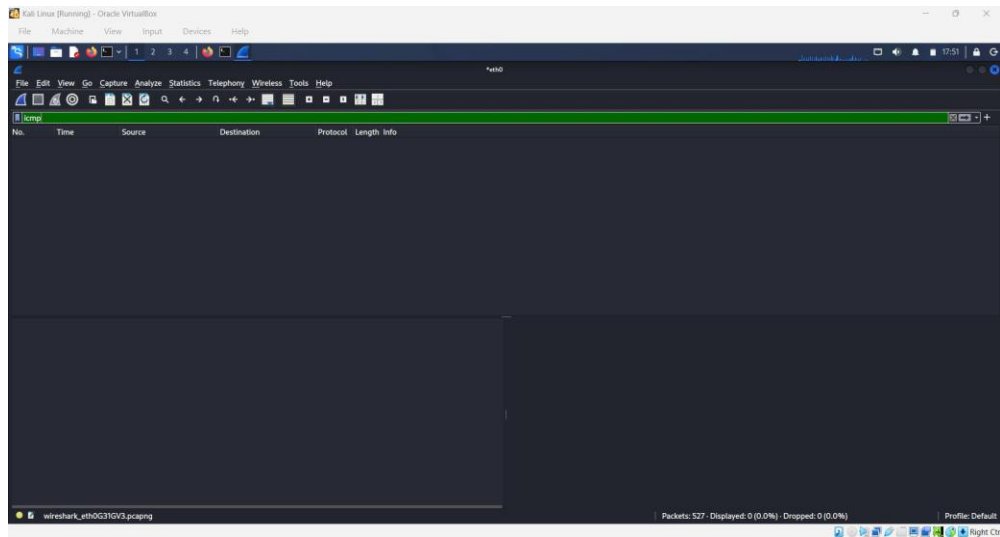
TCP : captures Wireshark running inside Kali Linux with the display filter set to `tcp`, isolating Transmission Control Protocol traffic over the `eth0` network interface. The packet list demonstrates the initial connection setup and encrypted session negotiation between the host machine at `10.0.2.4` and remote web services hosted at public IP addresses `34.149.226.178` and `199.232.105.91`.



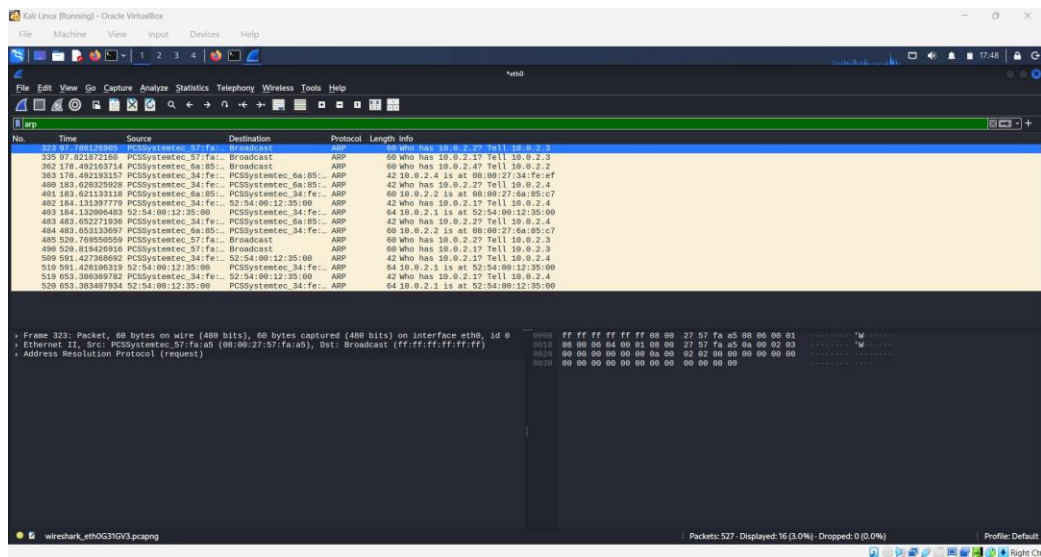
UDP : It shows Wireshark executing inside a Kali Linux environment with the display filter set to `udp`, isolating User Datagram Protocol traffic captured across the `eth0` interface. Because DNS relies on UDP as its underlying transport layer protocol for standard resolution, the displayed packet list consists entirely of Domain Name System queries and responses exchanged between host `10.0.2.4` and the upstream DNS resolver at `49.205.171.194`.



ICMP : It shows Wireshark running within a Kali Linux environment with the display filter set to `icmp`. As shown in the status bar at the bottom right, zero out of the total 527 captured packets match this criteria, resulting in a completely empty packet list pane.

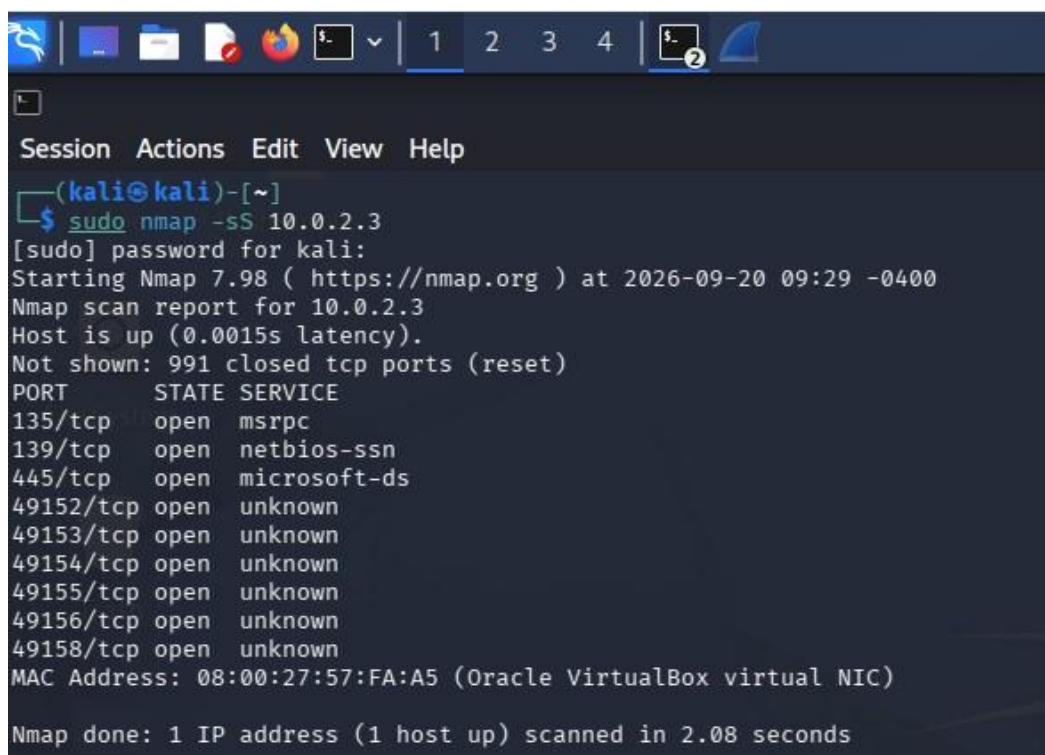


ARP : It shows Wireshark executing inside a Kali Linux virtual environment with the display filter set to `arp`, isolating Address Resolution Protocol traffic captured across the `eth0` interface. The packet list pane captures Layer 2 address resolution mechanics used to map IPv4 network addresses to local Media Access Control (MAC) hardware addresses within the local virtual network segment.



Task 10 – Nmap + Wireshark Investigation

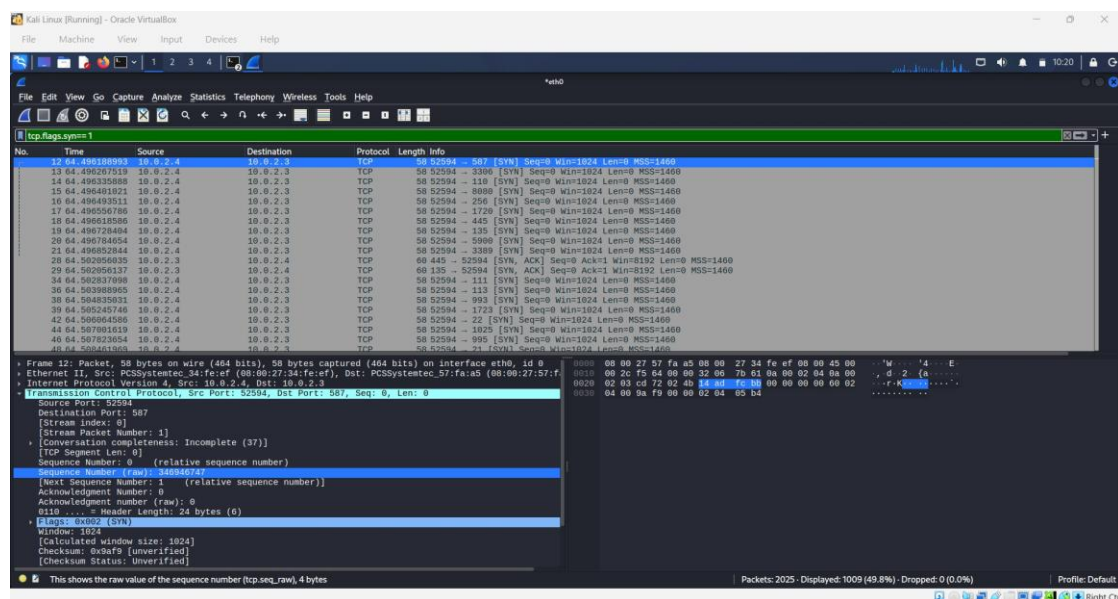
This screenshot shows a Kali Linux terminal window executing an Nmap TCP SYN stealth scan against target IP address `10.0.2.3`. The command `sudo nmap -sS 10.0.2.3` was issued, leveraging root privileges to craft raw SYN packets and evaluate target port states without establishing full TCP connections.



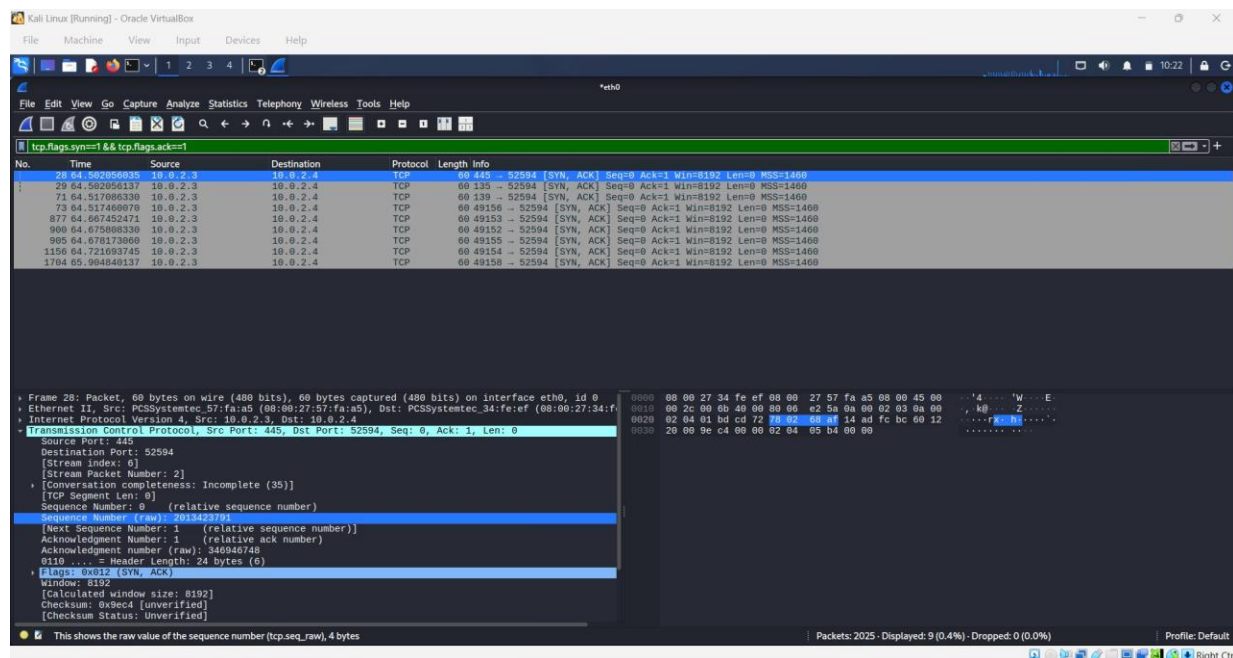
```
(kali@kali)-[~]
$ sudo nmap -sS 10.0.2.3
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 09:29 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0015s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
49158/tcp  open  unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

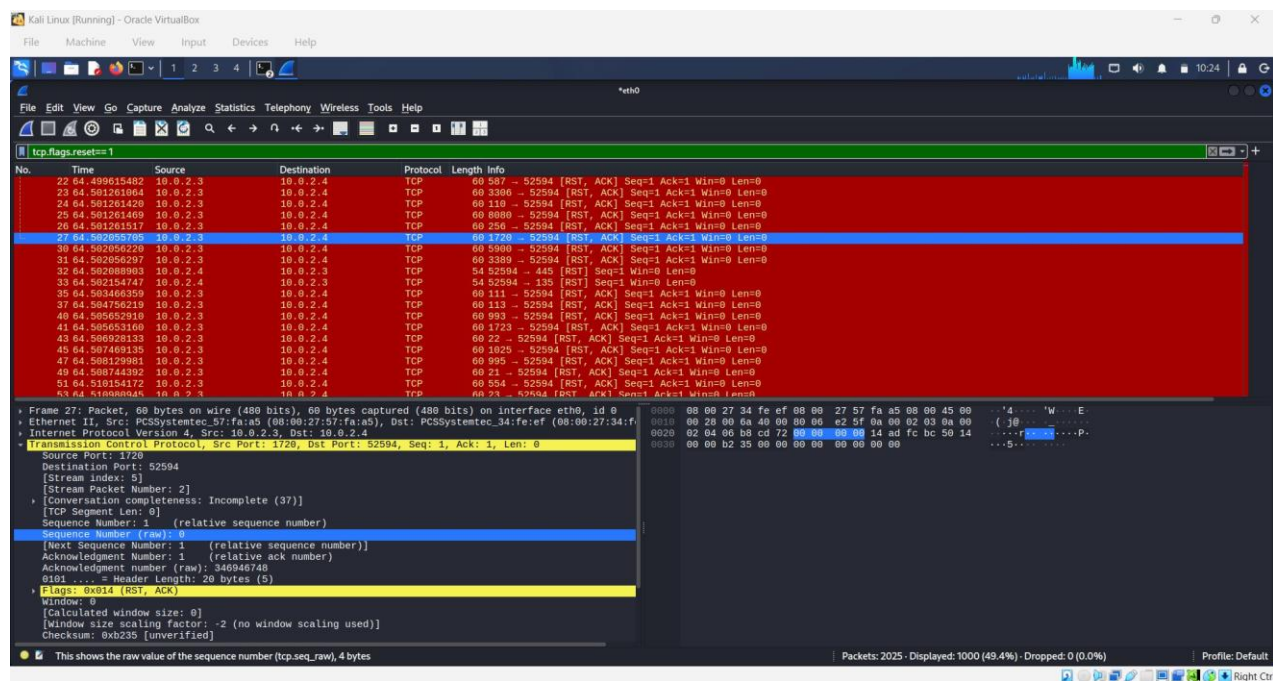
Nmap done: 1 IP address (1 host up) scanned in 2.08 seconds
```

TCP SYN : This screenshot shows Wireshark running on Kali Linux with the display filter set to `tcp.flags.syn == 1`. This filter isolates all TCP synchronization packets, capturing initial connection attempts across the `eth0` network interface.



This screenshot displays Wireshark running on Kali Linux with the display filter set to `tcp.flags.syn==1 && tcp.flags.ack==1`. This specific compound filter isolates TCP synchronization-acknowledgment packets, filtering out initial probes to capture only positive responses sent by open ports on the target host back to the scanning machine across interface `eth0`.





CONCLUSION : The Nmap SYN scan identified TCP services on the authorized Windows 7 VM. Wireshark provided packet-level evidence of the TCP connection attempts and responses. Comparing the two results demonstrated how SYN, SYN/ACK, and RST packets can be used to understand TCP port states during a network assessment.



Task 11 – Advanced Network Traffic Investigation

DNS : Capturing Wireshark running on a Kali Linux virtual machine with the display filter set to `dns`, effectively isolating Domain Name System query and response traffic over the `eth0` network interface. The traffic log clearly displays interaction between the client workstation at IP `10.0.2.4` and the upstream DNS resolver at `49.205.171.194`.

The screenshot shows the Wireshark interface on a Kali Linux virtual machine. The display filter is set to `dns`. The packet list shows a series of DNS queries and responses between the client workstation at IP `10.0.2.4` and the upstream DNS resolver at `49.205.171.194`. The packet details pane shows the structure of a DNS query for `content-signature-2.cdn.mozilla.net`.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.4	49.205.171.194	DNS	95	Standard query 0xabdd A content-signature-2.cdn.mozilla.net
2	0.00048524	10.0.2.4	49.205.171.194	DNS	95	Standard query 0x3bb AAAA content-signature-2.cdn.mozilla.net
3	0.000506222	10.0.2.4	49.205.171.194	DNS	75	Standard query 0x3a89 A ads.mozilla.org
4	0.011679693	10.0.2.4	49.205.171.194	DNS	75	Standard query 0xc6ff AAAA ads.mozilla.org
5	0.013579830	10.0.2.4	49.205.171.194	DNS	97	Standard query 0x5ff1 A firefox.settings.services.mozilla.com
6	0.013892411	10.0.2.4	49.205.171.194	DNS	97	Standard query 0xaccf AAAA firefox.settings.services.mozilla.com
7	0.014784663	49.205.171.194	10.0.2.4	DNS	111	Standard query response 0xabdd A content-signature-2.cdn.mozilla.net A 34.149.226.178
8	0.015642778	49.205.171.194	10.0.2.4	DNS	123	Standard query response 0x3bb AAAA content-signature-2.cdn.mozilla.net AAAA 2600:1901:0:8082::
12	0.019644694	49.205.171.194	10.0.2.4	DNS	127	Standard query response 0x3a89 A ads.mozilla.org CNAME mozilla.map.fastly.net A 199.232.105.91
13	0.021590389	49.205.171.194	10.0.2.4	DNS	139	Standard query response 0xc6ff AAAA ads.mozilla.org CNAME mozilla.map.fastly.net AAAA 2a04:de42:5a::347
14	0.022441943	49.205.171.194	10.0.2.4	DNS	149	Standard query response 0x5ff1 A firefox.settings.services.mozilla.com CNAME mozilla.map.fastly.net A 199.232.105.91
15	0.023650557	49.205.171.194	10.0.2.4	DNS	161	Standard query response 0xaccf AAAA firefox.settings.services.mozilla.com CNAME mozilla.map.fastly.net AAAA 2a04:de42:5a::347
130	0.621804721	10.0.2.4	49.205.171.194	DNS	85	Standard query 0x3910 A push.services.mozilla.com
131	0.622964219	10.0.2.4	49.205.171.194	DNS	85	Standard query 0x251e AAAA push.services.mozilla.com
136	0.628510453	49.205.171.194	10.0.2.4	DNS	161	Standard query response 0x3910 A push.services.mozilla.com A 34.167.243.93
137	0.629198408	49.205.171.194	10.0.2.4	DNS	160	Standard query response 0x251e AAAA push.services.mozilla.com SOA ns-679.awdns-20.net
211	1.709888282	10.0.2.4	49.205.171.194	DNS	71	Standard query 0xc23 A example.org
212	1.709499213	10.0.2.4	49.205.171.194	DNS	71	Standard query 0x472e A example.org
213	1.709873350	10.0.2.4	49.205.171.194	DNS	71	Standard query 0x302f AAAA example.org
214	1.710261071	10.0.2.4	49.205.171.194	DNS	78	Standard query 0xfbd4 HTTPS cloudflare-dns.com

Frame 1: Packet, 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface eth0, id 0

Ethernet II, Src: PCSysntec34:fe:ef (08:00:27:34:fe:ef), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)

Internet Protocol Version 4, Src: 10.0.2.4, Dst: 49.205.171.194

User Datagram Protocol, Src Port: 55288, Dst Port: 53

Domain Name System (query)

Transaction ID: 0xabdd

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

content-signature-2.cdn.mozilla.net: type A, class IN

Name: content-signature-2.cdn.mozilla.net

Name Length: 35

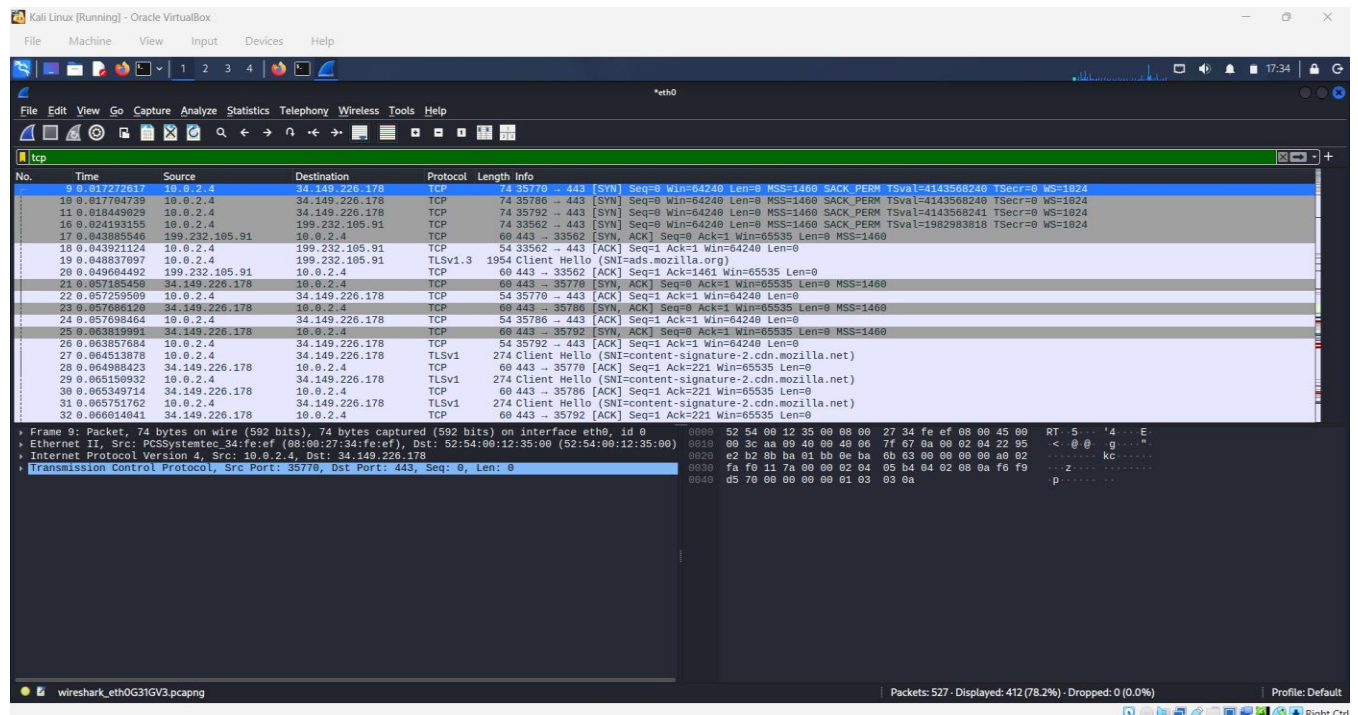
Label Count: 4

Type: A (1) (Host Address)

Class: IN (0x0001)

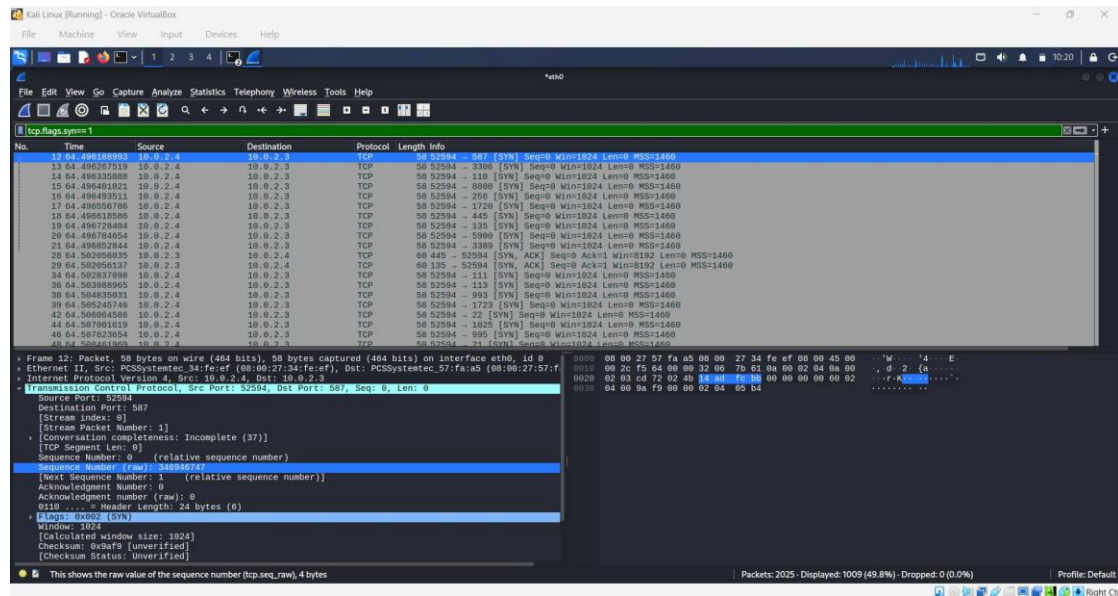
Response: IN (1)

TCP : captures Wireshark running inside Kali Linux with the display filter set to `tcp`, isolating Transmission Control Protocol traffic over the `eth0` network interface. The packet list demonstrates the initial connection setup and encrypted session negotiation between the host machine at `10.0.2.4` and remote web services hosted at public IP addresses `34.149.226.178` and `199.232.105.91`.



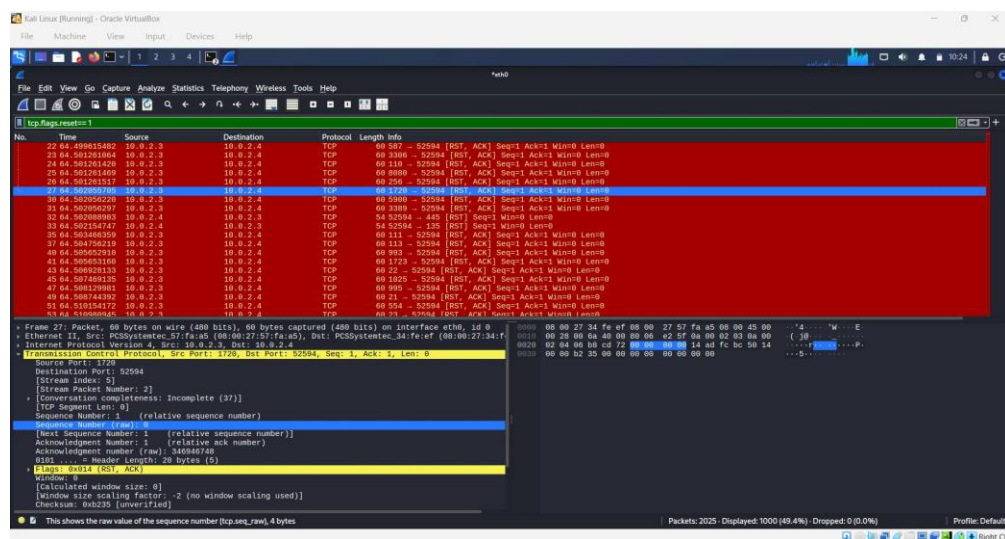
ICMP : It shows Wireshark running within a Kali Linux environment with the display filter set to `icmp`. As shown in the status bar at the bottom right, zero out of the total 527 captured packets match this criteria, resulting in a completely empty packet list pane.

TCP SYN : This screenshot shows Wireshark running on Kali Linux with the display filter set to `tcp.flags.syn == 1`. This filter isolates all TCP synchronization packets, capturing initial connection attempts across the `eth0` network interface.

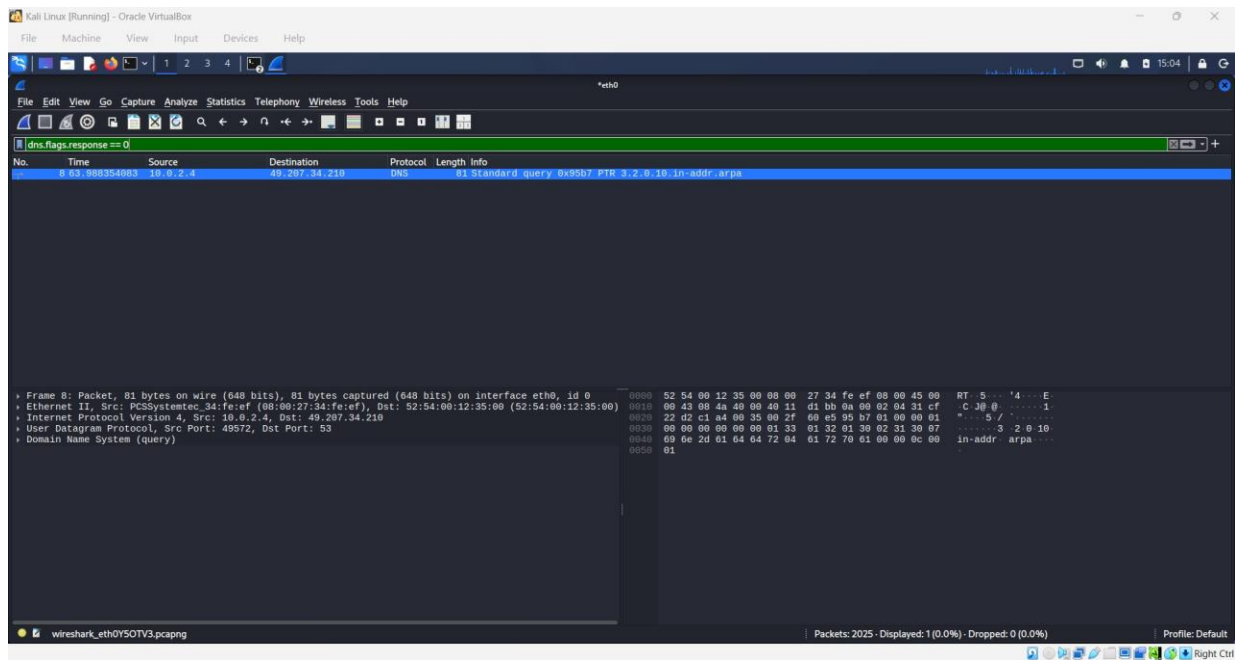


RST PACKETS : TCP RST packets were observed between the Kali Linux VM and the Windows 7 test VM during the network assessment.

Security significance : A TCP reset can indicate that a connection was rejected or abruptly terminated. During a port scan, RST responses can help determine that a TCP port is closed.



DNS queries only : DNS (Domain Name System) queries are requests made by a device to resolve a domain name into an IP address. In Wireshark, DNS query packets can be filtered to observe only the requests sent by clients.





TASK -12 Vulnerability Assessment

A vulnerability assessment is the process of identifying security weaknesses or exposures in a system and evaluating their potential security impact. In this lab, the assessment is based on the network discovery, Nmap scans, and Wireshark traffic analysis performed on the authorized Windows 7 virtual machine.

```
(kali@kali)-[~]
$ nmap -sV 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 15:39 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00092s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
49158/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)
Service Info: Host: WINDOWS7; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 59.96 seconds
```

A vulnerable SMBv1 service can create a significant security risk because an attacker may potentially exploit the affected service for remote code execution. Therefore, SMBv1 should be disabled where it is not required, and the operating system should be appropriately updated and protected.

```

(kali@kali)-[~]
$ nmap -p 135,139,445,49152,49153,49154,49155,49156,49158 --script vuln 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 15:52 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00060s latency).

PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
49158/tcp  open  unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Host script results:
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms17-010:
|   VULNERABLE:
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs: CVE:2017-0143
|     Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SMBv1
|       servers (ms17-010).
|
|     Disclosure date: 2017-03-14
|     References:
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|       https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 110.27 seconds

```

Findings	Evidence	Risk	Potential Impact	Recommendation
MS17-010 / SMBv1 vulnerability	NSE smb-vuln-ms17-010, CVE-2017-0143	High	Potential remote code execution	Apply appropriate Windows security updates and disable SMBv1 if not required
SMB/Microsoft-DS exposed	445/tcp open microsoft-ds	Medium	Increased SMB attack surface	Restrict access to trusted systems
NetBIOS-SSN exposed	139/tcp open netbios-ssn	Medium	Legacy service exposure	Disable/restrict if unnecessary
MSRPC exposed	135/tcp open msrpc	Medium	Increased RPC attack surface	Restrict RPC access
Multiple RPC ports exposed	49152–49158/tcp open	Medium	Additional exposed services	Restrict unnecessary RPC traffic

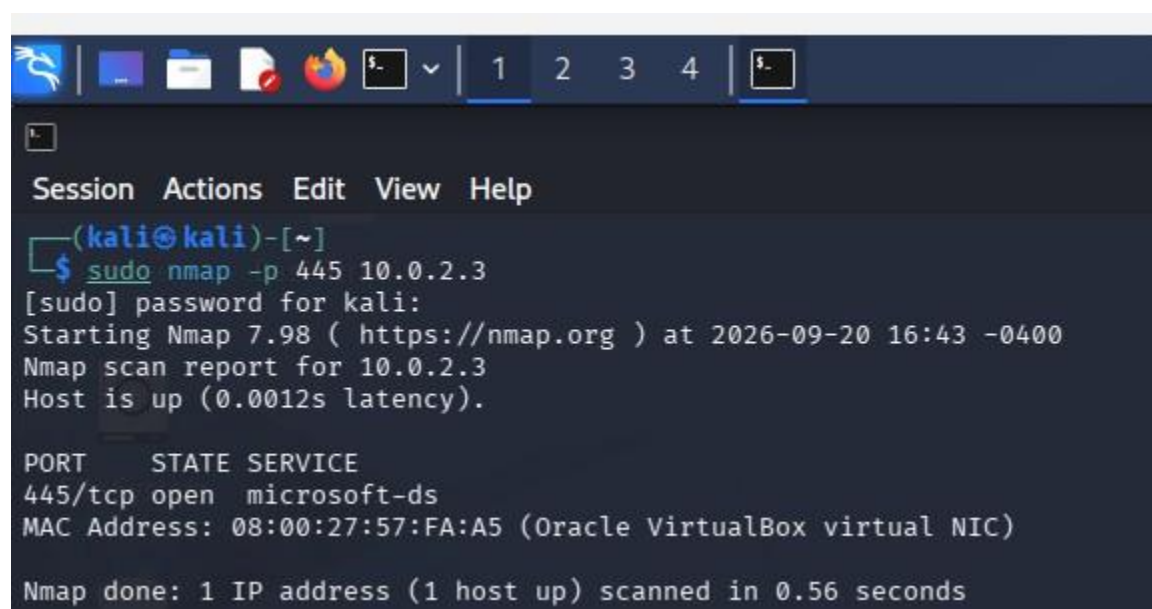


TASK - 13 Security Hardening & Before/After Testing

```
(kali㉿kali)-[~]
└─$ nmap -sV 10.0.2.3 PCSSystemer 34 fe ef (08:00:27:34 fe ef), Dst: 52:54:00:12:35:00 34.210
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 15:39 -0400
Nmap scan report for 10.0.2.3 Src Port: 49572, Dst Port: 53
Host is up (0.00092s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
49158/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)
Service Info: Host: WINDOWS7; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 59.96 seconds
```

BEFORE HARDENING :



```
(kali㉿kali)-[~]
└─$ sudo nmap -p 445 10.0.2.3
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 16:43 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0012s latency).

PORT      STATE SERVICE
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds
```

AFTER HARDENING :

```
unicastResponseToMulticast      Enable
Logging:
LogAllowedConnections          Disable
LogDroppedConnections          Disable
LogFileName                    %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                     4096

Public Profile Settings:
-----
State                           ON
Firewall Policy                 BlockInbound,AllowOutbound
LocalFirewallRules              N/A (GPO-store only)
LocalConSecRules               N/A (GPO-store only)
InboundUserNotification        Enable
RemoteManagement               Disable
unicastResponseToMulticast      Enable
Logging:
LogAllowedConnections          Disable
LogDroppedConnections          Disable
LogFileName                    %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                     4096
k.

C:\Windows\system32>
```

```
(kali㉿kali)-[~]
└─$ sudo nmap -p 445 10.0.2.3
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 16:59 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0011s latency).

PORT      STATE      SERVICE
445/tcp   filtered  microsoft-ds
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.76 seconds
```

Inbound Rules					
Name	Group	Profile	Enabled	Action	
Block NetBIOS TCP 139		All	Yes	Block	
Block SMB TCP 445		All	Yes	Block	
Core Networking - Destination Unreacha...	Core Networking	All	Yes	Allow	
Core Networking - Destination Unreacha...	Core Networking	All	Yes	Allow	
Core Networking - Dynamic Host Config...	Core Networking	All	Yes	Allow	
Core Networking - Dynamic Host Config...	Core Networking	All	Yes	Allow	
Core Networking - Internet Group Mana...	Core Networking	All	Yes	Allow	
Core Networking - IPHTTPS (TCP-In)	Core Networking	All	Yes	Allow	
Core Networking - IPv6 (IPv6-In)	Core Networking	All	Yes	Allow	
Core Networking - Multicast Listener Do...	Core Networking	All	Yes	Allow	
Core Networking - Multicast Listener Qu...	Core Networking	All	Yes	Allow	
Core Networking - Multicast Listener Rep...	Core Networking	All	Yes	Allow	
Core Networking - Multicast Listener Rep...	Core Networking	All	Yes	Allow	
Core Networking - Neighbor Discovery A...	Core Networking	All	Yes	Allow	
Core Networking - Neighbor Discovery S...	Core Networking	All	Yes	Allow	
Core Networking - Packet Too Big (ICMP...	Core Networking	All	Yes	Allow	
Core Networking - Parameter Problem (I...	Core Networking	All	Yes	Allow	
Core Networking - Router Advertisement...	Core Networking	All	Yes	Allow	
Core Networking - Router Solicitation (IC...	Core Networking	All	Yes	Allow	
Core Networking - Teredo (UDP-In)	Core Networking	All	Yes	Allow	
Core Networking - Time Exceeded (ICMP...	Core Networking	All	Yes	Allow	
Distributed Transaction Coordinator (RPC)	Distributed Transaction Coo...	Domain	No	Allow	
Distributed Transaction Coordinator (RPC)	Distributed Transaction Coo...	Private...	No	Allow	
Distributed Transaction Coordinator (RP...	Distributed Transaction Coo...	Domain	No	Allow	
Distributed Transaction Coordinator (RP...	Distributed Transaction Coo...	Private...	No	Allow	
Distributed Transaction Coordinator (TC...	Distributed Transaction Coo...	Domain	No	Allow	
Distributed Transaction Coordinator (TC...	Distributed Transaction Coo...	Private...	No	Allow	
File and Printer Sharing (Echo Request - I...	File and Printer Sharing	Private...	No	Allow	
File and Printer Sharing (Echo Request - I...	File and Printer Sharing	Domain	No	Allow	
File and Printer Sharing (Echo Request - I...	File and Printer Sharing	Domain	No	Allow	
File and Printer Sharing (Echo Request - I...	File and Printer Sharing	Private...	No	Allow	
File and Printer Sharing (LLMNR-UDP-In)	File and Printer Sharing	All	No	Allow	

```

(kali@kali)-[~]
$ sudo nmap -p 139 10.0.2.3
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 17:16 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0013s latency).

PORT      STATE      SERVICE
139/tcp   filtered  netbios-ssn
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds

```

```

(kali@kali)-[~]
└─$ sudo nmap -p 49152-49158 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 17:33 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0015s latency).

PORT      STATE      SERVICE
49152/tcp  filtered
49153/tcp  filtered
49154/tcp  filtered
49155/tcp  filtered
49156/tcp  filtered
49157/tcp  filtered
49158/tcp  filtered
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.78 seconds

```

Hardening Action	Before	Action Taken	After	Improvement
Block TCP 445	445/tcp open	Created firewall rule to block TCP 445	445/tcp filtered	Reduced SMB exposure
Block TCP 139	139/tcp open	Created firewall rule to block TCP 139	139/tcp filtered	Reduced NetBIOS exposure
Block RPC high ports	49152–49158/tcp open	Created firewall rule to block TCP 49152–49158	49152–49158/tcp filtered	Reduced exposure of dynamic RPC ports



TASK - 14 Final Mini Security Assessment

The objective of the final assessment was to evaluate the security posture of the authorized Windows 7 test VM after network discovery, service enumeration, traffic analysis, vulnerability assessment, and security hardening.

```
(kali@kali)-[~]
$ sudo nmap -p 49152-49158 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 17:33 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0015s latency).

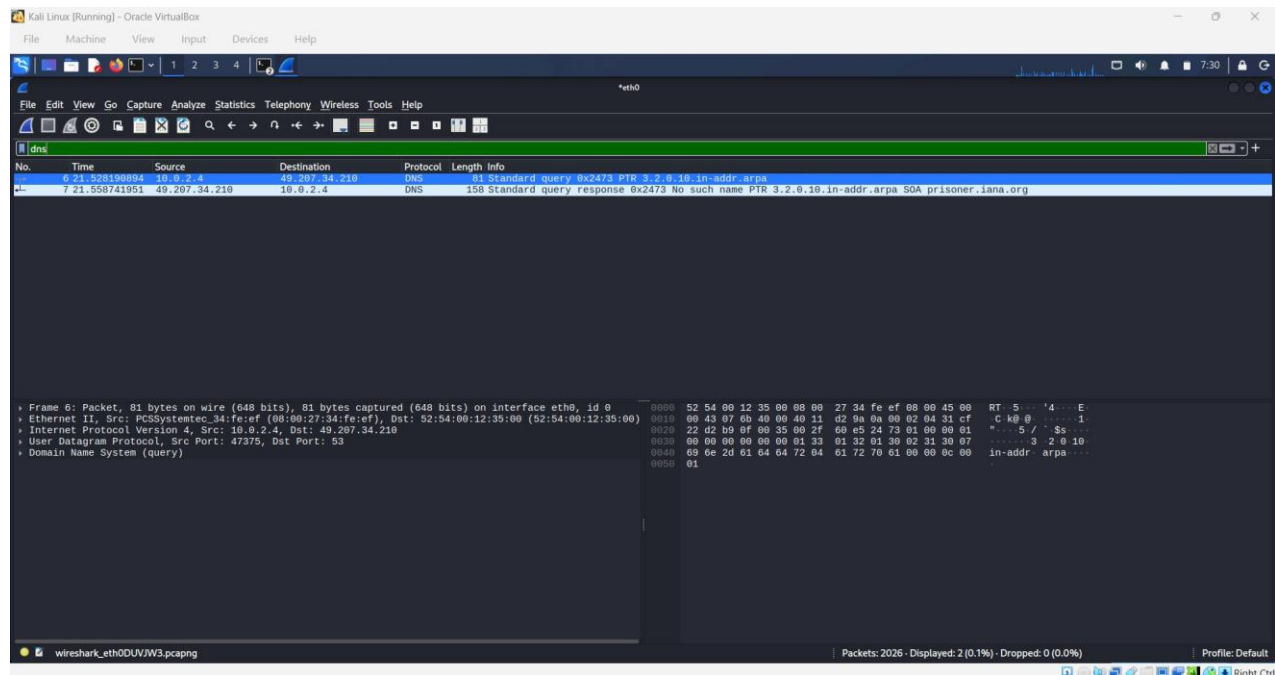
PORT      STATE      SERVICE
49152/tcp  filtered   unknown
49153/tcp  filtered   unknown
49154/tcp  filtered   unknown
49155/tcp  filtered   unknown
49156/tcp  filtered   unknown
49157/tcp  filtered   unknown
49158/tcp  filtered   unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.78 seconds
```

2. Before vs. After

Port	Before Hardening	After Hardening
445/TCP	Open	Filtered
139/TCP	Open	Filtered
49152–49158 /TCP	Open	Filtered

A final Wireshark DNS capture confirmed that the target's normal DNS resolution and network connectivity continued to function correctly after the firewall changes were applied, with no adverse impact on ordinary operation.



Conclusion

This Week 3 assessment carried a full offensive-to-defensive workflow:

- identifying the target
- enumerating services with Nmap
- validating behavior at the packet level with Wireshark
- correlating scan traffic to captured packets
- translating findings into concrete hardening actions

Overall, the assessment strengthened practical skills across reconnaissance, traffic analysis, and defensive hardening within a safe, authorized lab environment.

THANK YOU